PH.45130 **FALL 2026**

Quantum Gravitation

Problem Set 6

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Topic

Propagators and Medium Causality

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Propagators and Medium Causality. State all assumptions and conventions.
$$(\partial_t^2 - c_m^2\nabla^2 + m_g^2 c_m^4/\hbar^2)G_R = \delta^4(x)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.
$$D_{\mu\nu,\rho\sigma} = \Pi^{(2)}_{\mu\nu,\rho\sigma}/(k^2 - m_g^2 + i\varepsilon)$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.
$$S_{obs} = K_{det} * G_R * J$$

Show enough intermediate work that another student can reproduce the calculation.